

The Lateral/Caudal Ganglionic Eminence Makes a Limited Contribution to Cortical Oligodendrocytes

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Jialin Li, Feihong Yang, Yu Tian, Ziwu Wang, Dashi Qi, Zhengang Yang, Jiangang Song, Jing Ding, Xin Wang , Zhuangzhi Zhang 

State Key Laboratory of Medical Neurobiology and MOE Frontiers Center for Brain Science, Institutes of Brain Science, and Department of Neurology, Zhongshan Hospital, Fudan University, Shanghai 200032, P.R. China • Department of Anesthesiology, Shuguang Hospital Affiliated with Shanghai University of Traditional Chinese Medicine, Shanghai 201203, P.R. China • Center for Clinical Research and Translational Medicine, Yangpu Hospital, School of Medicine, Tongji University, Shanghai 200090, P.R. China

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Abstract

The emergence of myelinating oligodendrocytes represents a pivotal developmental milestone in vertebrates, given their capacity to ensheath axons and facilitate the swift conduction of action potentials. It is widely accepted that cortical oligodendrocyte progenitor cells (OPCs) arise from medial ganglionic eminence (MGE), lateral/caudal ganglionic eminence (LGE/CGE) and cortical radial glial cells (RGCs). Here, we used two different fate mapping strategies to challenge the established notion that the LGE generates cortical OPCs. Furthermore, we used a Cre/loxP-dependent exclusion strategy to reveal that the LGE/CGE-derived OPCs are minimal. Additionally, we showed that specifically eliminating MGE-derived OPCs leads to a significant reduction of cortical OPCs. Together, our findings indicate that the contribution of OPCs from LGE/CGE is minimal, contrary to previous beliefs. These findings provide a new view of the developmental origins of cortical OPCs and a valuable foundation for future research on both normal development and oligodendrocyte-related disease.

eLife assessment

The authors provide **solid** evidence that any contribution of oligodendrocyte precursors to the developing cortex from the lateral ganglionic eminence is minimal in scope. The methods used support the conclusions, with some technical concerns that the authors can address with further experimentation. These are considered **valuable** additions to our understanding of the origins of oligodendrocytes in the forebrain during development.

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Introduction

Oligodendrocytes (OLs) are one of the most abundant cell types in the mammalian neocortex. The emergence of myelination OLs in vertebrates is a major event in the evolution of the central nervous system (CNS), as these cells play a crucial role in enhancing axonal insulation and the conduction of neural impulses. Previously, they have been described as support cells that myelinate axons and play essential roles in the function of neocortical circuits. Myelin loss or dysfunction is associated with many disorders, such as multiple sclerosis, depression and pediatric leukodystrophies^{1,2,3}.

During CNS development, OLs originate from oligodendrocyte precursor cells (OPCs), which arise from different regions of the embryonic neural tube. OPCs proliferate and migrate away from the ventricular germinal zone (VZ) into developing gray and white matter before differentiating into mature OLs^{3,4}. The developmental origin of cortical oligodendrocytes has been extensively studied in the CNS, especially in the neocortex^{5–8}. A landmark study⁹ demonstrated that cortical OPCs are generated in three waves, starting with a ventral wave, gradually transitioning to more dorsal origins. The first wave of cortical OPCs is derived from the medial ganglionic eminence (MGE) or the anterior entopeduncular area (AEP) at embryonic day E12.5 (*Nkx2.1*⁺ lineage). The second wave of cortical OLs is derived from the lateral/caudal ganglionic eminences (LGE/CGE) at E15.5 (*Gsx2*⁺/*Nkx2.1*[−] lineage). The final wave occurs at P0, when OPCs originate from cortical glial progenitor cells (*Emx1*⁺ lineage). Our recent studies^{10,11}, however, challenge the notion that cortical OLs are generated in these three waves^{9,12–14}, as the cortical glial progenitor cells were also found to express *Gsx2*, which was known to be exclusively expressed in the developing subpallium^{15–17}, after E16.5¹¹. Therefore, *Gsx2*⁺/*Nkx2.1*[−] lineage OLs may not represent LGE-derived cortical OLs. To date, whether the LGE/CGE generates cortical OPCs (OLs) during cortical development remains an unanswered question in the field of developmental neuroscience.

The present study was undertaken to answer this question. We carried out distinct fate mapping experiments and intersectional lineage analyses to reveal that cortical OPCs do not derive from the LGE/CGE. First, using in utero electroporation (IUE) to target the LGE in a Cre recombinase-dependent *IS* (*Rosa*^{CAG-LSLFlrt-tdTomato-Frt-EGFP}) reporter mouse line¹⁸, we found that LGE-derived OPCs produced striatal OLs rather than cortical OLs. Second, by combining the PiggyBac transposon system with IUE, we found that no traced cells migrated into the cortex from the LGE. Third, by crossing the *Nkx2.1*^{Cre} and *Emx1*^{Cre} mouse lines with an *Rosa*^{H2B-GFP} reporter mouse line, we found that nearly all cortical OPCs originated from *Nkx2.1*^{Cre} or *Emx1*^{Cre} lineage cells. Finally, we analyzed *Olig2*-NCKO (*Nkx2.1*^{Cre}; *Olig2*^{F/F}) mice, and found that the number of cortical OPCs was significantly reduced from E14.5 to E16.5. These results indicated that the MGE was the predominant ventral source of cortical OPCs. Taken together, our findings suggest that the LGE/CGE gives rise to limited cortical OPCs (OLs) during neocortical development.

Results

Fate mapping of LGE-derived OPCs by combining IUE with a Cre recombinase-dependent *IS* reporter

OPCs initially emerge in the MGE and can be labeled by platelet-derived growth factor receptor (α-subunit, *Pdgfra*) around E12.5. We hypothesized that if the LGE generated cortical OPCs, the *Pdgfra*⁺ cell migration stream from the LGE to the cortex could be observed during cortical development. To confirm this speculation, we conducted a detailed analysis of *Pdgfra* mRNA

expression using *in situ* hybridization (ISH). Our results showed that there was no observable *Pdgfra*⁺ cell migration stream from the LGE to the cortex from E13.5 to P0 (**Figure 1A**). Next, we used Cre recombinases in combination with an *IS* reporter mouse line¹⁸ to fate map LGE-derived OPCs (OLs). We delivered pCAG-Cre plasmids specifically into the LGE VZ of *IS* reporter mouse embryos at E13.5 by IUE (**Figures 1B-C**). In this experiment, cells generated from electroporated neural stem cells (NSCs) expressed tdTomato (tdT). Interestingly, we observed many tdT⁺ cells throughout the whole LGE (striatum) at E18.5. In contrast, there were very few tdT⁺ cells detected in the cortex (**Figure 1D**). OPCs express *Olig2*, *Sox10*, *Pdgfra* and NG2 (*Cspg4*) and mature neurons express NeuN^{19–25}. To determine the identity of the tdT⁺ cells, we performed double staining of tdT with OLIG2, SOX10, PDGFRA, NG2 (CSPG4) and NeuN. Within the cortex, tdT⁺ cells did not express OPC marker genes but express NeuN; however, tdT⁺ cells expressed OPC marker genes and NeuN in the striatum (**Figure 1E-F**). We also delivered pCAG-Cre plasmids specifically into the LGE VZ of *IS* reporter mice at P0 by IUE and analyzed these mice at P49 (**Figure 1G**) to investigate whether the LGE generates cortical OLs at a later developmental stage. Our results showed that LGE progenitors generate striatum OLs and astrocytes, which expressed SOX10 and SOX9, respectively (**Figure 1G**).

To further determine whether LGE-derived OPCs migrate into the cortex after E18.5, we used the same strategy as mentioned above (Figure 1-figure supplement 1A). We observed that at P10, the majority of the tdT⁺ cells were located in the striatum, whereas very few tdT⁺ cells were located in the neocortex (Figure 1-figure supplement 1B). All tdT⁺ cells in the neocortex expressed NeuN but did not express OLIG2 (Figure 1-figure supplement 1C). In fact, tdT⁺ cells in the neocortex did not express OPC marker genes; instead, they expressed NeuN in the neocortex. In contrast, tdT⁺ cells in the striatum expressed OPC marker genes (*Olig2*, *Sox10* and *Pdgfra*) and NeuN (Figure 1-figure supplement 1D), consistent with our observation at E18.5 (**Figure 1E**). We also found that tdT⁺ cells expressed the astrocyte marker ALDH1L1 in the striatum at P10 (Figure 1-figure supplement 1D). Furthermore, through consecutive tissue sectioning and 3D reconstruction, we revealed that the traced cells tended to be localized in the striatum (Figure 1-figure supplement 1F). Collectively, our findings indicated that the LGE gives rise to striatal OLs, neurons and astrocytes, but not cortical OLs.

Lineage tracing of LGE-derived OPCs by combining IUE with the PiggyBac transposon system

To further investigate whether the LGE generates cortical OPCs during cortical development, we employed an alternative fate mapping strategy by combining IUE with the PiggyBac transposon system to map the fate of OPCs derived from LGE. This approach utilized a transposase enzyme (helper plasmid) to integrate the H2B-GFP sequence from a donor plasmid into the genome of the in utero electroporated LGE progenitors along the lateral ventricular zone. The H2B-GFP reporter vector was delivered specifically into the LGE VZ of wild-type embryos at E13.5 by IUE, and these embryos were examined at E18.5 or P10 (**Figures 2A-B**). In this system, most traced cells (GFP⁺) were located in the striatum, and very few traced cells were observed in the cortex (**Figures 2C and E**). Usually, a small population of RGCs located in the dorsal LGE or lateral/ventral pallium was labeled by IUE. These RGCs mainly migrated to the lateral/ventral pallium and amygdala (**Figure 2C**). We analyzed 139 GFP⁺ cells in the cortex (N = 4), but we did not observe any GFP⁺ cells that were OPCs in the dorsal cortex, as these cells did not express OLIG2, SOX10 or PDGFRA (**Figure 2D**). Instead, they expressed NeuN (**Figure 2D**). In contrast, GFP⁺ cells expressed OLIG2, SOX10, PDGFRA and NeuN in the striatum at E18.5 (**Figure 2D**). The same results were also observed at P10 (**Figure 2E**). Again, our findings strongly suggested that the LGE may not generate cortical OPCs.

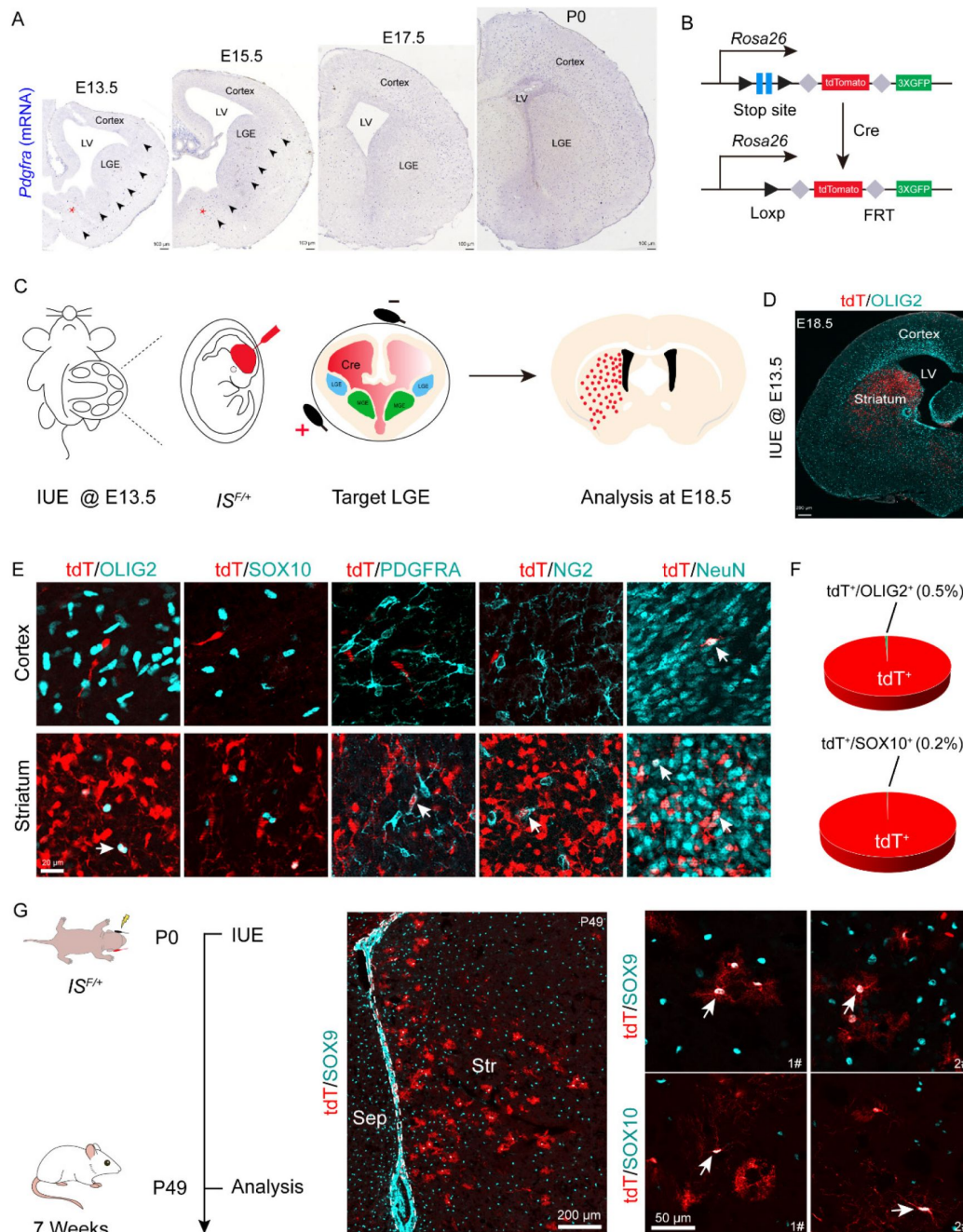


Figure 1.

Fate mapping of LGE-derived OPCs by combining IUE with a Cre recombinase-dependent IS reporter

- (A) *In situ* hybridization showing *Pdgfra* expression from E13.5 to P0. Arrowheads indicated the *Pdgfra*⁺ cell migration stream from the MGE/AEP to the cortex.
- (B) Scheme of the IS reporter lines.
- (C) Experimental schedule to trace LGE-derived OPCs.
- (D) Representative coronal sections showing the distribution of tdT⁺ cells.
- (E) tdT⁺ cells expressed NeuN but not OLIG2, SOX10, PDGFRA or NG2 in the cortex. In contrast, the tdT⁺ cells expressed all of these markers (OLIG2, SOX10, PDGFRA, NG2 and NeuN) in the striatum.
- (F) The ratio of the OLIG2⁺ and SOX10⁺ cells of the electroporated cells in the cortex. N = 4 mice per group.
- (G) Fate mapping of LGE-derived cells at P0. tdT⁺ cells expressed SOX10 and SOX9 in the striatum at P49.

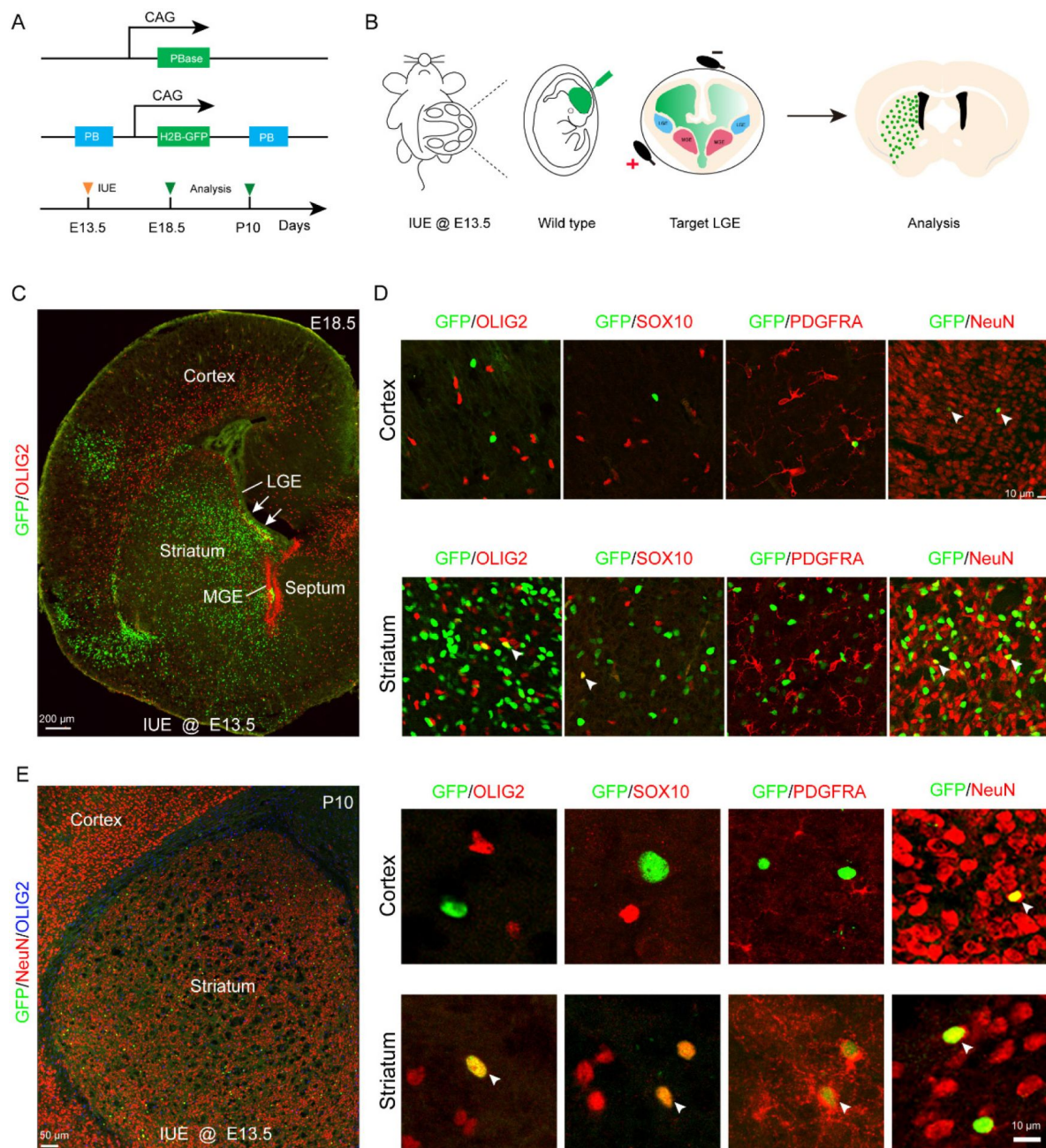


Figure 2.

Fate mapping of LGE-derived OPCs by combining IUE with PiggyBac transposon system

- (A) Schematic of the PiggyBac transposon reporter system used in this study.
- (B) The experimental workflow.
- (C) Representative coronal sections showing the distribution of GFP⁺ cells at E18.5.
- (D) GFP⁺ cells expressed NeuN but not glial cell markers, such as OLIG2, SOX10 and PDGFRA in the cortex. However, GFP⁺ cells expressed NeuN, OLIG2, SOX10 and PDGFRA in the striatum.
- (E) GFP⁺ cells expressed NeuN but not glial cell markers, such as OLIG2, SOX10 and PDGFRA in the cortex and GFP⁺ cells expressed NeuN, OLIG2, SOX10 and PDGFRA in the striatum at P10.

An exclusion strategy revealed that LGE/CGE contribution to cortical OPCs is minimal

A previous report⁹ suggested that cortical OPCs originate from MGE, LGE/CGE and cortical RGCs. When MGE-derived cortical OPCs were eliminated, LGE/CGE-derived cells accounted for approximately 50% of cortical OPCs at P10. Subsequently, another study reported that approximately 20% OLs in corpus callosum are ventral-derived at P12/P13²⁶. However, according to our recent findings^{10,11}, we speculate that these OPCs (*Gsx2*⁺ lineage) may derived from the cortical progenitors (*Emx1*⁺ lineage). To confirm these findings, using the *Emx1*^{Cre}; H2B-GFP mice, we found that many GSX2⁺ cells were located in the cortical SVZ. As expected, these GSX2⁺ cells were co-labelled with GFP (*Emx1*⁺ lineage), suggesting that cortical SVZ GSX2⁺ cells were derived from cortical progenitors (Figure 3-figure supplement 1A-C).

While distinct fate mapping strategies provided evidence that LGE may not generate cortical OPCs during cortical development, there remained a possibility that a small population of LGE/CGE RGCs or progenitors might generate a subpopulation of cortical OPCs at specific time points. To address this possibility, we crossed the *Emx1*^{Cre 27–29}, *Nkx2.1*^{Cre 30} and *Rosa*^{H2B-GFP18} mouse lines to generate *Emx1*^{Cre}; *Nkx2.1*^{Cre}; *Rosa*^{H2B-GFP} mice. Because *Emx1*^{Cre} was specifically expressed in cortical RGCs²⁷ and *Nkx2.1*^{Cre} was specifically expressed in MGE/AEP RGCs³⁰, cortical OPCs were traced (GFP⁺ cells) in using *Emx1*^{Cre}; *Nkx2.1*^{Cre}; *Rosa*^{H2B-GFP} mice. GFP⁺ cell OPCs, on the other hand, were considered as LGE/CGE-derived cortical OPCs (OLs) (Figures 3A–B). We performed double staining of GFP with OPC marker genes, such as *Sox10* and *Pdgfra* in *Emx1*^{Cre}; *Nkx2.1*^{Cre}; H2B-GFP mice (Figure 3C). We observed that the majority of OPCs or OLs were co-labeled with GFP in *Emx1*^{Cre}; *Nkx2.1*^{Cre}; *Rosa*^{H2B-GFP} mice at P10 (Figure 3C). Statistical analysis showed that only approximately 2% of OPCs were GFP⁺ cells, and they were possibly derived from the LGE/CGE (Figures 3D–E), which was consistent with our observation at P0 (Figures 3G–I). Importantly, the expression of *Nkx2.1*^{Cre} is lower within the dorsalmost region of the MGE than in other *Nkx2.1*-expressing regions³⁰. Thus, these GFP⁺ OPCs (~2%) may be derived from the dorsal MGE rather than from the LGE/CGE. Using this approach, we revealed that nearly all cortical OLs/OPCs were derived from MGE/AEP and cortical RGCs. Taken together, our findings indicated that the contribution of cortical OPCs from the LGE/CGE is minimal.

The MGE/AEP was the predominant ventral source of the cortical OPCs

Olig2, a basic helix-loop-helix transcription factor, plays a critical role in the regulation of OPC fate determination. To investigate the role of *Olig2* in the MGE, we crossed *Nkx2.1*^{Cre} mice with *Olig2*^{F/+} mice to eliminate *Olig2* expression in the SVZ/VZ of the MGE/AEP (Figures 4A–B). We referred to the conditional mutants as *Olig2*-NCKO mice. A previous study⁹ showed that MGE/AEP-derived OPCs migrate into the cortex around E14.5, and LGE/CGE-derived OPCs migrate into cortex around E15.5. We assumed that the MGE/AEP was the predominant ventral source of cortical OPCs, and that inhibiting MGE/AEP-derived OPC generation would therefore significantly reduce the number of cortical OPCs at E16.5. To test this possibility, we analyzed *Olig2*-NCKO mice, in which the generation of MGE/AEP-derived OPCs was inhibited. Our results showed that the expression of OPC marker genes, such as *Sox10* and *Pdgfra* was significantly reduced (> 10-fold) in the cortex of *Olig2*-NCKO mice, compared to the control mice at E14.5 and E16.5 (Figures 4C–E). Considering the recombination efficiency of the CRE enzyme, the limited presence of residual OPCs in the cortex of *Olig2*-NCKO mice may be arise from the MGE/AEP. Indeed, we observed few PDGFRA-positive cells in the MGE/AEP of the *Olig2*-NCKO mice (Figure 4F). Altogether, our findings indicated that the MGE/AEP was the predominant ventral source of cortical OPCs.

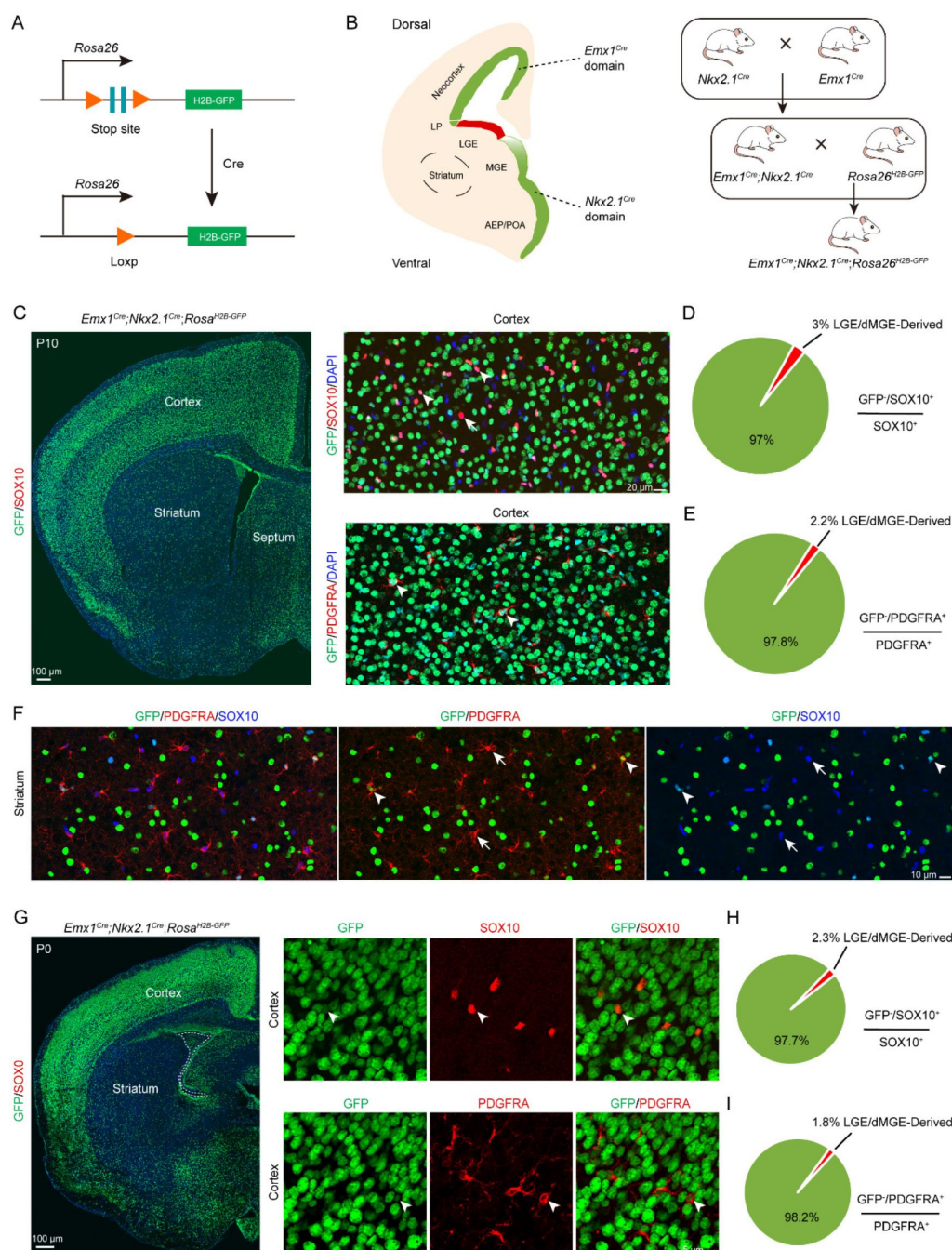


Figure 3.

An exclusion strategy showed that LGE/CGE contribution to cortical OPCs is minimal

(A) Scheme of the H2B-GFP reporter lines.

(B) Experimental design of the exclusion strategy to trace the lineage of LGE RGCs.

(C) Representative coronal sections showing the traced cells in the forebrain.

The majority of SOX10- and PDGFRA-positive cells were GFP⁺ cells in the cortex at P10.

(D-E) The pie chart shows that the percentage of LGE/dMGE-derived cortical OPCs was approximately 3%. N = 6 mice per group.

(F) Many GFP⁺ cells expressed SOX10 and PDGFRA in the striatum at P10.

(G) Nearly all SOX10 and PDGFRA expressed GFP in the cortex at P0.

(H-I) The pie chart shows that the percentage of LGE/dMGE-derived cortical OPCs was less than 3%. N = 6 mice per group.

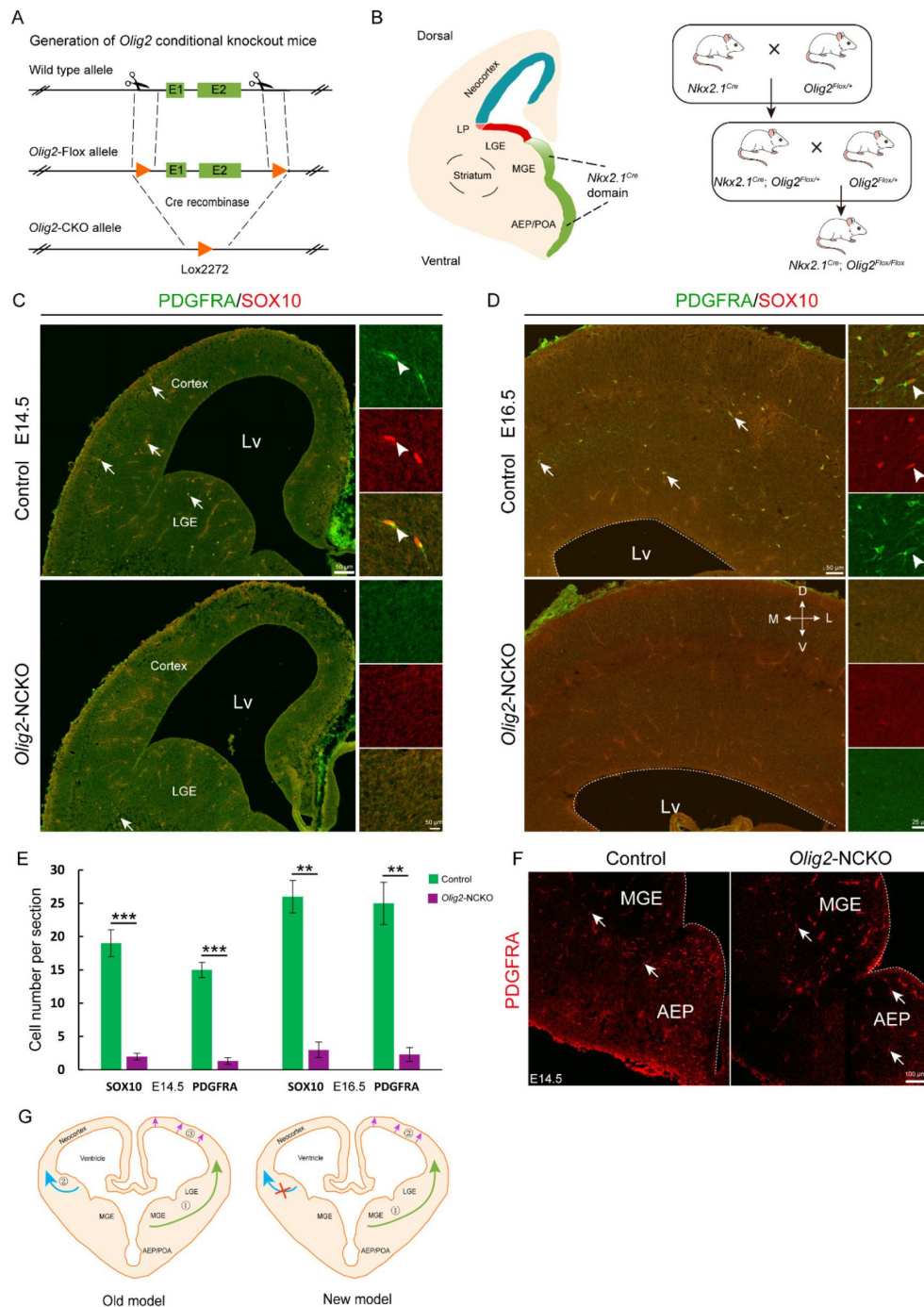


Figure 4.

The MGE may be the sole ventral source of cortical OPCs

- (A) The CRISPR/Cas9 technique was used to generate an *Olig2* conditional knockout allele.
- (B) Experimental design for the generation of *Olig2*-NCKO mice.
- (C-D) The expression of SOX10 and PDGFRA was significantly reduced in the cortex of *Olig2*-NCKO mice compared with that of control mice.
- (E) The number of SOX10- and PDGFRA-positive cells was significantly reduced in *Olig2*-NCKO mice compared with control mice. Student's t-test, ** $P < .01$, *** $P < .001$, $n \geq 5$ mice per group, mean $\pm$ SEM.
- (F) Few PDGFRA-positive cells were detected in the MGE/AEP of the *Olig2*-NCKO mice.
- (G) A new model of the developmental origins of cortical OPCs.

Discussion

It is widely accepted in the field of developmental neuroscience that cortical OPCs originate from three regions: the MGE/AEP, LGE/CGE and cortex. However, in the current study, we re-examined the developmental origins of cortical OPCs using diverse strategies. Our findings show that cortical OPCs may not come from the LGE/CGE, contrary to previous reports^{9,12}, which suggested that the LGE/CGE generates cortical OPCs (**Figure 4G**). In fact, no cortical OPCs migrating from the LGE to the cortex were observed via our two fate mapping strategies. Furthermore, an exclusion strategy suggested that limited contribution of the LGE/CGE to cortical OPCs. In addition, according to our study of *Olig2*-NCKO mice, MGE/AEP seems to be the predominant ventral source of cortical OPCs during neocortical development. Overall, these findings provide strongly evidence that the contribution of the cortical OPCs from the LGE/CGE is minimal in the mouse brain.

Why LGE is not the main OPC pool for the cortex during development

OLs are specialized cells of the CNS that produce myelin, a multilayered membrane that spirally enwraps axons and facilitates rapid nerve conduction. Recent studies have shown that cortical OLs are heterogeneous, as there are transcriptionally distinct subpopulations of cortical OLs during development and disease^{8,31–33}. Indeed, OLs are derived from distinct OPC pools, which acquire different abilities to respond to demyelination^{9,34}. In the spinal cord, the developmental origin of OLs has been hotly debated for many years. Some groups believed that OLs originate only from the ventral neural tube, whereas others tend to support multiple origins^{12,35–37}. One pioneering study⁹ purported that, in the forebrain, cortical OLs derive from multiple origins, such as the MGE, LGE and cortex.

In general, the specification of OLs depends on SHH protein released from the notochord and floor plate, as *Shh* expression is both necessary and sufficient for OL generation in the spinal cord³⁸. As in the spinal cord, OL generation in the cortex is also dependent on the *Shh* signaling pathway³⁹. During MGE development, *Nkx2.1* and *Six3* promote SHH secretion. Furthermore, the early-born neurons of the MGE also secrete SHH protein before E14.5⁴⁰. Thus, the high levels of SHH in the MGE contribute to the generation of OPCs from RGCs. In contrast, *Shh* signaling pathway activity is very weak in the LGE at the early stage (before E14.5). As the level of SHH gradually increases in the LGE, this results in a shift from neurogenesis to gliogenesis after E15.5, and the large number of the striatal neurons may block the migration of OPCs into the neocortex^{41,42}. In summary, the process of cortical OL generation can be summarized as follows: First, the high level of SHH in the MGE allows for the production of a small population of cortical OPCs around E12.5. Subsequently, multipotent intermediate progenitors begin to express DLX transcription factors resulting in ending the generation of OPCs in the MGE^{43–45}. Second, radial glial cells (RGCs) undergo a gradual transition from neurogenesis to gliogenesis around E15.5, driven by the progressive elevation of SHH levels in the LGE. However, the LGE had already generated a substantial number of neurons during this period, blocking the migration of OPCs into the cortex. Finally, with the gradual increase of SHH concentration, cortical RGCs begin to cease neurogenesis and shift toward gliogenesis after E16.5^{10,41,46}. Therefore, the generation of cortical OPCs is regulated by both precise temporal and spatial patterning, which is strictly dependent on the local environment.

Cortical-derived OLs/OPCs make up the majority of the cortical OLs/OPCs in the adult brain

During the early development of the cortex, both pallium and subpallium can generate cortical OPCs^{9,47}. Interestingly, most of the ventral-derived OPCs, particularly MGE-derived OPCs/OLs, are eventually eliminated due to the continued expansion of dorsal-derived OPCs in the adult cortex^{6,9}. It appears that OPCs generated later in development may possess a competitive advantage over their earlier-generated counterparts. Indeed, human glial progenitor cells (hGPCs) engrafted neonatally or transplanted into the adult mouse brain can effectively disperse throughout the forebrain, differentiate into mature OLs and remyelinate the demyelinated adult brain^{1,48}. These findings indicate that OPCs generated later in development have the ability to take over for earlier-generated OPCs in the adult brain.

Based on the present work and other studies^{8,9,47}, cortical-derived OPCs probably constitute the majority of cortical OPCs, including mature cortical OLs. Thus, OLs may function as a homogeneous population in the adult forebrain. In fact, the number of LGE/CGE-derived cortical OPCs may also gradually decrease in the adult brain, as reported by Lui et al⁶. However, Lui et al. did not analyze LGE-derived and MGE-derived OPCs separately. Therefore, they did not exclude the possibility that the LGE/CGE does not generate cortical OPCs. In summary, cortical OPCs/OLs comprise MGE-derived and cortical RGC-derived OPCs/OLs during the neonatal stage. In contrast, cortical-derived OPCs/OLs make up the majority of the cortical OPCs/OLs in the adult cortex. Recently, another study showed that ventral-derived OLs contribution to adult cortex is minimal⁴⁹.

In summary, our study provides strong evidence that the contribution of the LGE to cortical OPCs is minimal during forebrain development. Thus, our findings provide new evidence about the developmental origins of cortical OPCs, which may guide future research on both normal development and disease.

Methods

Animals

All experiments conducted in this study were in accordance with the guidelines of Fudan University. We used *Emx1*^{Cre 27,28}, and *Nkx2.1*^{Cre 30} mice, in conjunction with either *Rosa*^{H2B-GFP18} or *Rosa-LS*¹⁸ for genetic fate mapping. *Olig2*^{F/+} mice were generated via the CRISPR/Cas9 strategy. Loxp sites flanked the coding region of exons 1 and 2. Wild-type mice or littermates without the Cre allele were used as controls. All mice were maintained on a mixed genetic background (C57BL/6J and CD1). The mice were allowed access to water and food ad libitum and maintained on a 12-hour light/dark cycle. The day of vaginal plug detection was considered embryonic day 0.5 (E0.5), and the day of birth was defined as postnatal day 0 (P0). Both male and female mice were used in all experiments.

Tissue preparation

Postnatal mice were deeply anesthetized and perfused intracardially thoroughly with ice-cold PBS followed by 4% paraformaldehyde (PFA). Embryos were harvested from deeply anesthetized pregnant mice. All brains were fixed overnight in 4% PFA at 4°C, cryoprotected in 30% sucrose for at least 24 hours, embedded in optimal cutting temperature (O.C.T.) (Sakura Finetek) on dry ice and ethanol slush and preserved at -80°C. All frozen mouse tissues were sectioned into 20 µm thick sections unless specifically mentioned and stained on glass slides.

Plasmid Construction

The pCAG-Cre vector was constructed as reported elsewhere. The coding sequence of Cre in the pCAG-Cre vector was replaced by the PiggyBac transposase to construct the pCAG-PiggyBac transposase vector. The coding DNA sequence (CDS) of Cre in the pCAG-Cre vector was replaced by H2B-GFP from the *Rosa*-H2B-GFP mouse strain to construct the PB-pCAG-H2B-GFP vector. The inverted terminal repeat (ITR) sequence of the PiggyBac transposon was cloned upstream of the CAG promoter and downstream of the beta-globin polyA sequence for recognition by the PiggyBac transposase.

In utero electroporation (IUE)

In utero electroporation (IUE) of wild-type or IS embryos was performed at E13.5. The pCAG-Cre (Addgene #13775), pCAG-GFP (Addgene #11150), pCAG-PiggyBac transposase or PB-pCAG-H2B-GFP (final concentration: 1–2 µg/µL) plasmid was mixed with 0.05% Fast Green (Sigma) and injected into the lateral ventricle of embryos (0.5 µL per embryo) using a beveled glass micropipette. Five electrical pulses (duration: 50 ms) were applied at 42 V across the uterine wall with a 950 ms interval between pulses. Electroporation was performed using a pair of 5 mm platinum electrodes (BTX, Tweezertrode 45-0488, Harvard Apparatus) connected to an electroporator (BTX, ECM830). The embryos were analyzed at E18.5 and P10.

Immunofluorescence

The sections were first washed with 0.05 M TBS for 10 min, incubated in Triton-X-100 (0.5% in 0.05 M TBS) for 30 min at room temperature (RT), and then incubated with blocking solution (5% donkey serum +0.5% Triton-X-100 in 0.05 M TBS, pH= 7.2) for 2 h at RT. The primary antibodies were diluted in 5% donkey serum, incubated overnight at 4°C, and then rinsed 3 times with 0.05 M TBS.

Secondary antibodies (1:500, all from Jackson ImmunoResearch) matching the appropriate species were added and incubated for 2–4 h at RT. Fluorescently stained sections were then washed 3 times with 0.05 M TBS for 10 min. The sections were then stained with 4',6-diamidino-2-phenylindole (DAPI) (1:3000, Sigma) diluted in TBS for 1 min and then finally rinsed three more times with TBS. The sections were then coverslipped with Gel/Mount (Biomed, Foster City, CA). The primary antibodies used in this study are shown in the key resource table.

In situ hybridization (ISH)

ISH was performed on 20-µm cryosections using digoxigenin riboprobes as previously described⁵⁰. The sections were first postfixed in 4% PFA for 20 min. mRNA *in situ* hybridization was performed as previously described⁴⁹. The probes were constructed from cDNA isolated from the wild-type mouse brain and amplified by PCR using the following primers:

Probe Name	Primer Sequence
<i>Pdgfra</i>	Primer F: TGGCCAGCAATGTCTCAAAT
	Primer R: TCCTTTCAAGCATGGGGACA

3D Reconstruction

Images of serial coronal sections along the rostral–caudal axis were collected via an Olympus VS 120 microscope using a 10X objective and used by Amira as a formwork for reconstruction. The coordinates of each labeled cell (tdT⁺) in the brain were determined. A stereotaxic mouse brain atlas was used to locate the brain anatomical structures where the labeled cells were positioned.

Image acquisition and statistical analysis

Images for quantitative analyses were acquired with an Olympus VS 120 microscope using a 10X objective. Bright-field images were acquired with an Olympus VS 120 microscope using a 10X objective. Other fluorescence images were taken with the Olympus FV3000 confocal microscope system using 20× or 40× objectives. The images were merged, cropped and optimized in Adobe Photoshop and Adobe Illustrator without distorting the original visual information.

Analyses were performed using Microsoft Excel. Unpaired t test was used to determine statistical significance. All quantitative data are presented as the mean ± SEM. Differences with *p* values < 0.05 were considered significant. For quantification of SOX10⁺, PDGFRA⁺ and GFP⁺ cells in the cortex at E18.5 and P10, six anatomically matched 20-μm thick coronal sections were selected (n = 6 mice per group). We counted SOX10⁺, PDGFRA⁺ and GFP⁺ cells in the cortex under a 20X objective lens. The data are presented as the number of SOX10⁺, PDGFRA⁺ and GFP⁺ cells per section for each cortex.

For quantification of SOX10⁺ and PDGFRA⁺ cells in the dorsal cortex at E14.5 and E16.5, five or six anatomically matched 20-μm thick coronal sections were selected (n = 6, E14.5; n=5, E16.5). We counted SOX10⁺ and PDGFRA⁺ cells in the cortex under a 20X objective lens. The data are presented as the number of SOX10⁺ and PDGFRA⁺ cells per section for each cortex.

Key resource table

Reagent Resource	or	Source	Identifier
Antibodies			
Chicken Anti-GFP		Aves Labs	GFP-1020, 1:3000
Rabbit Anti-ALDH1L1		Abcam	ab87117, 1:1000

Rabbit Anti- OLIG2	Oasis Biofarm	OB-PRB009, 1:500
Rat Anti-OLIG2	Oasis Biofarm	OB-PRT020, 1:500
Mouse Anti-OLIG2	Millipore	MABN50, 1:1000
Rabbit Anti- OLIG2	Millipore	AB9610, 1:1000
Goat Anti-SOX10	R&D Systems	ab442208, 1:500
Guinea Pig Anti-SOX10	Oasis Biofarm	OB-PGP001, 1:1000
Rabbit Anti-PDGFRα	Oasis Biofarm	OB-PRB051, 1:800
Rabbit Anti-NeuN	Bioscience	R-3770-100, 1:1000
Rabbit Anti-NeuN	Oasis Biofarm	OB-PRB039, 1:1000
Guinea Pig Anti-NG2	Oasis Biofarm	OB-PGP002, 1:1000

Experimental models

Mouse: C57BL/6	Department of Laboratory Animal Science at Fudan University	http://10.107.12.196/
Mouse: CD1	Department of Laboratory Animal Science at Fudan University	http://10.107.12.196/

Mouse: <i>IS</i> reporter	The Jackson Laboratory	Stock No. 028582
Mouse: <i>Olig2^{F/+}</i>	This manuscript	N/A; available from the authors
Mouse: <i>Nkx2.1^{Cre}</i>	The Jackson Laboratory	Stock No. 008661
Mouse: <i>Emx1^{Cre}</i>	The Jackson Laboratory	Stock No. 005628

Recombinant DNA

pCAGIG	Addgene	Cat#11159
pGL4.10	Promega	Cat#E6651
pGL4.23	Promega	Cat#E8411

All secondary antibodies were from Jackson ImmunoResearch Labs.

Secondary Antibodies	Cat Number	Dilution
Alexa®488-Conjugated Affinipure Donkey Anti-Chicken IgY ⁺⁺ (IgG) (H ⁺ L)	703-545-155	1:500
Cy TM 3-Conjugated Affinipure Donkey Anti-Rabbit IgG (H ⁺ L)	711-165-152	1:500
Alexa®647-Conjugated Affinipure Donkey Anti-Rat IgG (H+L)	712-605-153	1:500
Alexa®647-Conjugated Affinipure Donkey Anti-Goat IgG (H ⁺ L)	705-605-147	1:500

Alexa®488-Conjugated Affinipure Donkey Anti-Rabbit IgG (H ⁺ L)	711-545-152	1:500
Cy TM 3-AffiniPure Donkey Anti-Goat IgG (H ⁺ L)	705-505-147	1:500
Cy TM 3-AffiniPure Donkey Anti-Rat IgG (H ⁺ L)	712-165-150	1:500
Cy TM 3-AffiniPure Donkey Anti-Guinea Pig IgG (H ⁺ L)	706-165-148	1:500

Acknowledgements

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Data availability statement

In this study, no dataset was utilized. However, any supplementary information necessary for the reanalysis of the data presented in this paper is obtainable from the corresponding author upon request.

Potential conflicts of interest

We declare that there is no conflict of interest.

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Editors

Reviewing Editor

Samuel Pleasure

University of California, San Francisco, San Francisco, United States of America

Senior Editor

Kathryn Cheah

University of Hong Kong, Hong Kong, Hong Kong

Reviewer #1 (Public Review):

Summary:

In this manuscript the authors re-examine the developmental origin of cortical oligodendrocyte (OL) lineage cells using a combination of strategies, focussing on the question of whether the LGE generates cortical OL cells. The paper is interesting to myelin biologists, the methods used are appropriate and, in general, the study is well-executed, thorough, and persuasive, but not 100% convincing.

Strengths, weaknesses, and recommendations:

The first evidence presented that the LGE does not generate OLs for the cortex is that there are no OL precursors 'streaming' from the LGE during embryogenesis, unlike the MGE (Figure 1A). This in itself is not strong evidence, as they might be more dispersed. In fact, in the images shown, there is no obvious 'streaming' from the MGE either. Note that in Figure 1 there is no reference to the star that is shown in the figure.

The authors then electroporate a reporter into the LGE at E13.5 and examine the fate of the electroporated cells (Figures 1C-E). They find that electroporated cells became neurons in the striatum and in the cortex but no OLs for the cortex. There are two issues with this: first, there is no quantification, which means there might indeed be a small contribution from the LGE that is not immediately obvious from snapshot images. Second, it is unexpected to find labelled neurons in the cortex at all since the LGE does not normally generate neurons for the cortex! Electroporations are quite crude experiments as targeting is imprecise and variable and not always discernible at later stages. For example, in Figure 1D, one can see tdTOM+ cells near the AEP, as well as the striatum. Hence, IUE cannot on its own be taken as proof that there is no contribution of the LGE to the cortical OL population.

The authors then use an alternative fate-mapping approach, again with E13.5 electroporations (Figure 2). They find only a few GFP+ cells in the cortex at E18 (Figures 2C-D) and P10 (Figure 2E) and these are mainly neurons, not OL lineage cells. Again, there is no quantification.

Figure 3 is more convincing, but the experiments are incomplete. Here the authors generate triple-transgenic mice expressing Cre in the cortex (Emx1-Cre) and the MGE (Nkx2.1-Cre) as well as a strong nuclear reporter (H2B-GFP). They find that at P0 and P10, 97-98% of OL-lineage cells (SOX10+ or PDGFRA+) in the cortex are labelled with GFP (Figure 3). This is a more convincing argument that the LGE/CGE might not contribute significant numbers of OL lineage cells to the cortex, in contrast to the Kessaris et al. (2006) paper, which showed that Gsh2-Cre mice label ~50% of SOX10+ve cells in the motor cortex at P10. The authors of the present paper suggest that the discrepancy between their study and that of Kessaris et al. (2006) is based on the authors' previous observation (Zhang et al 2020) (<https://doi.org/10.1016/j.celrep.2020.03.027>) that GSH2 is expressed in intermediate precursors of the cortex from E18 onwards. If correct, then Kessaris et al. might have mistakenly attributed Gsh2-Cre+ lineages to the LGE/CGE when they were in fact intrinsic to the cortex. However, the evidence from Zhang et al 2020 that GSH2 is expressed by cortical intermediate precursors seems to rest solely on their location within the developing cortex; a more convincing demonstration would be to show that the GSH2+ putative cortical precursors co-label for EMX1 (by immunohistochemistry or in situ hybridization), or that they co-label with a reporter in Emx1-driven reporter mice. This demonstration should be simple for the authors as they have all the necessary reagents to hand. Without these additional data, the assertion that GSX2+ve cells in the cortex are derived from the cortical VZ relies partly on an act of faith on the part of the reader.

Note that Tripathi et al. (2011, "Dorsally- and ventrally-derived oligodendrocytes have similar electrical properties but myelinate preferred tracts." J. Neurosci. 31, 6809-6819) found that the Gsh-Cre+ OL lineage contributed only ~20% of OLs to the mature cortex, not ~50% as reported by Kessaris et al. (2006). If it is correct that these Gsh2-derived OLs are from the cortical anlagen as the current paper claims, then it would raise the possibility that the ventricular precursors of GSH2+ intermediate progenitors are not uniformly distributed through the cortical VZ but are perhaps localized to some part of it. Then the contribution of Gsh2-derived OLs to the cortical population could depend on precisely where one looks relative to that localized source. It would be a nice addition to the current manuscript if the authors could explore the distribution of their GSH2+ intermediate precursors throughout the developing cortex. In any case, Tripathi et al. (2011) should be cited.

Finally, the authors deleted *Olig2* in the MGE and found a dramatic reduction of PDGFRA+ and SOX10+ cells in the cortex at E14 and E16 (Figure 4A-F). This further supports their conclusion that, at least at E16, there is no significant contribution of OLs from ventral sources other than the MGE/AEP. This does not exclude the possibility that the LGE/CGE generates OLs for the cortex at later stages. Hence, on its own, this is not completely convincing evidence that the LGE generates no OL lineage cells for the cortex.

Comments on the latest version:

The revised manuscript has addressed the issues we raised previously. The addition of the new Figure 3 supplement 1A-C demonstrating that *Gsx2*+ve cells in the cortex are generated from *Emx1*-Cre precursors is convincing, although there is nothing to prove that the GFP+, *Gsh2*+ double-labelled nuclei are oligodendrocyte lineage and not, for example, astrocytes. It would be helpful to include a *Gsh2*, *Olig2* (or *Gsh2*, *Sox10*) double-label image to prove this point. Also, to make the figure more clear, the authors should also show a small area at high magnification, splitting the green and red channels so that the reader can see more clearly that all the red cells are also green.

<https://doi.org/10.7554/eLife.94317.2.sa2>

Reviewer #2 (Public Review):

Traditional thinking has been that cortical oligodendrocyte progenitor cells (OPCs) arise in the development of the brain from the medial ganglionic eminence (MGE), lateral/caudal ganglionic eminence (LGE/CGE), and cortical radial glial cells (RGCs). Indeed a landmark study demonstrated some time ago that cortical OPCs are generated in three waves, starting with a ventral wave derived from the medial ganglionic eminence (MGE) or the anterior entopeduncular area (AEP) at embryonic day E12.5 (*Nkx2.1*+ lineage), followed by a second wave of cortical OLs derived from the lateral/caudal ganglionic eminences (LGE/CGE) at E15.5 (*Gsx2*/*Nkx2.1*- lineage), and then a final wave occurring at P0, when OPCs originate from cortical glial progenitor cells (*Emx1*+ lineage). However, the authors challenge the idea in this paper that cortical progenitors are produced from the LGE. They have found previously that cortical glial progenitor cells were also found to express *Gsx2*, suggesting this may not have been the best marker for LGE-derived OPCs. They have used fate mapping experiments and lineage analyses to suggest that cortical OPCs do not derive from the LGE.

Strengths:

(1) The data is high quality and very well presented, and experiments are thoughtful and elegant to address the questions being raised.

(2) The authors use two elegant approaches to lineage trace LGE derived cells, namely fate mapping of LGE-derived OPCs by combining IUE (intrauterine electroporation) with a Cre recombinase-dependent IS reporter, and Lineage tracing of LGE-derived OPCs by combining IUE with the PiggyBac transposon system. Both approaches show convincingly that labelled LGE-derived cells that enter the cortex do not express OPC markers, but that those co-labelling with oligodendrocyte markers remain in the striatum.

(3) The authors then use further approaches to confirm their findings. Firstly they lineage trace *Emx1*-Cre; *Nkx2.1*-Cre; H2B-GFP mice. *Emx1*-Cre is expressed in cortical RGCs and *Nkx2.1*-Cre is specifically expressed in MGE/AEP RGCs. They find that close to 98% of OPCs in the cortex co-label with GFP at later times, suggesting the contribution of OPCs from LGE is minimal.

(4) They use one further approach to strengthen the findings yet further. They cross Nkx2.1-Cre mice with Olig2 F/+ mice to eliminate Olig2 expression in the SVZ/VZ of the MGE/AEP (Figures 4A-B). The generation of MGE/AEP-derived OPCs is inhibited in these Olig2-NCKO conditional mice. They find that the number of cortical progenitors at E16.5 is reduced 10-fold in these mice, suggesting that LGE contribution to cortical OPCs is minimal.

Impact of Study:

The authors show elegantly and convincingly that the contribution of the LGE to the pool of cortical OPCs is minimal. The title should perhaps be that the LGE contribution is minimal rather than no contribution at all, as they are not able to rule out some small contribution from the LGE. These findings challenge the traditional belief that the LGE contributes to the pool of cortical OPCs. The authors do show that the LGE does produce OPCs, but that they tend to remain in the striatum rather than migrate into the cortex. It is interesting to wonder why their migration patterns may be different from the MGE-derived OPCs which migrate to the cortex. The functional significance of these different sources of OPCs for adult cortex in homeostatic or disease states remains unclear though.

<https://doi.org/10.7554/eLife.94317.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

In this manuscript the authors re-examine the developmental origin of cortical oligodendrocyte (OL) lineage cells using a combination of strategies, focussing on the question of whether the LGE generates cortical OL cells. The paper is interesting to myelin biologists, the methods used are appropriate and, in general, the study is well-executed, thorough, and persuasive, but not 100% convincing.

Thank you very much for approving our paper.

Strengths, weaknesses, and recommendations:

The first evidence presented that the LGE does not generate OLs for the cortex is that there are no OL precursors 'streaming' from the LGE during embryogenesis, unlike the MGE (Figure 1A). This in itself is not strong evidence, as they might be more dispersed. In fact, in the images shown, there is no obvious 'streaming' from the MGE either. Note that in Figure 1 there is no reference to the star that is shown in the figure.

We totally agree with you. While OPC migration stream is not strong evidence to support that the LGE does not generate OPCs for the cortex, when considering our additional evidence, the absence of obvious 'streaming' from LGE to cortex provided supplementary support for this conclusion. Finally, we have removed the star in the figure.

The authors then electroporate a reporter into the LGE at E13.5 and examine the fate of the electroporated cells (Figures 1C-E). They find that electroporated cells became neurons in the striatum and in the cortex but no OLs for the cortex. There are two issues with this: first, there is no quantification, which means there might indeed be a small contribution from the LGE that is not immediately obvious from snapshot images.

Second, it is unexpected to find labelled neurons in the cortex at all since the LGE does not normally generate neurons for the cortex. Electroporations are quite crude experiments as targeting is imprecise and variable and not always discernible at later stages. For example, in Figure 1D, one can see tdTOM+ cells near the AEP, as well as the striatum. Hence, IUE cannot on its own be taken as proof that there is no contribution of the LGE to the cortical OL population.

Thank you for your constructive suggestions.

(1) Following the reviewer's suggestion, we have added these statistics, please see Figure 1F.

(2) The reviewer raised a good point. We occasionally found a very small number of electroporated cells in the MGE/AEP VZ in our IUE system. Therefore, we can identify these electroporated cells in the cortex, most of them expressed the neuronal marker NeuN. We suspect these are MGE-derived cortical interneurons. It's worth noting that these electroporated cells (MGE-derived) are not glia cells. The probable reason may be that MGE/AEP generate cortical OPCs mainly before E13.5 (in this study we performed IUE at E13.5).

The authors then use an alternative fate-mapping approach, again with E13.5 electroporations (Figure 2). They find only a few GFP+ cells in the cortex at E18 (Figures 2C-D) and P10 (Figure 2E) and these are mainly neurons, not OL lineage cells. Again, there is no quantification.

Thank you very much for your suggestions. Actually, in this fate-mapping approach, the electroporated cells in the cortex is very few. We analyzed four mice, and found that all GFP positive cells (139 GFP+) did not express OLIG2, SOX10 and PDGFRA.

Figure 3 is more convincing, but the experiments are incomplete. Here the authors generate triple-transgenic mice expressing Cre in the cortex (Emx1-Cre) and the MGE (Nkx2.1-Cre) as well as a strong nuclear reporter (H2B-GFP). They find that at P0 and P10, 97-98% of OL-lineage cells (SOX10+ or PDGFRA+) in the cortex are labelled with GFP (Figure 3). This is a more convincing argument that the LGE/CGE might not contribute significant numbers of OL lineage cells to the cortex, in contrast to the Kessaris et al. (2006) paper, which showed that Gsh2-Cre mice label ~50% of SOX10+ve cells in the motor cortex at P10. The authors of the present paper suggest that the discrepancy between their study and that of Kessaris et al. (2006) is based on the authors' previous observation (Zhang et al 2020) (<https://doi.org/10.1016/j.celrep.2020.03.027>) that GSH2 is expressed in intermediate precursors of the cortex from E18 onwards. If correct, then Kessaris et al. might have mistakenly attributed Gsh2-Cre+ lineages to the LGE/CGE when they were in fact intrinsic to the cortex. However, the evidence from Zhang et al 2020 that GSH2 is expressed by cortical intermediate precursors seems to rest solely on their location within the developing cortex; a more convincing demonstration would be to show that the GSH2+ putative cortical precursors co-label for EMX1 (by immunohistochemistry or in situ hybridization), or that they co-label with a reporter in Emx1-driven reporter mice. This demonstration should be simple for the authors as they have all the necessary reagents to hand. Without these additional data, the assertion that GSX2+ve cells in the cortex are derived from the cortical VZ relies partly on an act of faith on the part of the reader. Note that Tripathi et al. (2011, "Dorsally- and ventrally-derived oligodendrocytes have similar electrical properties but myelinate preferred tracts." J. Neurosci. 31, 6809-6819) found that the Gsh-Cre+ OL lineage contributed only ~20% of OLs to the mature cortex, not ~50% as reported by Kessaris et al. (2006). If it is correct that these Gsh2-derived OLs are from the cortical anlagen as the current paper claims, then it would raise the possibility that the ventricular precursors of GSH2+ intermediate progenitors are not uniformly distributed through the cortical VZ but are

perhaps localized to some part of it. Then the contribution of Gsh2-derived OLs to the cortical population could depend on precisely where one looks relative to that localized source. It would be a nice addition to the current manuscript if the authors could explore the distribution of their GSH2+ intermediate precursors throughout the developing cortex. In any case, Tripathi et al. (2011) should be cited.

Thank you for your constructive suggestions.

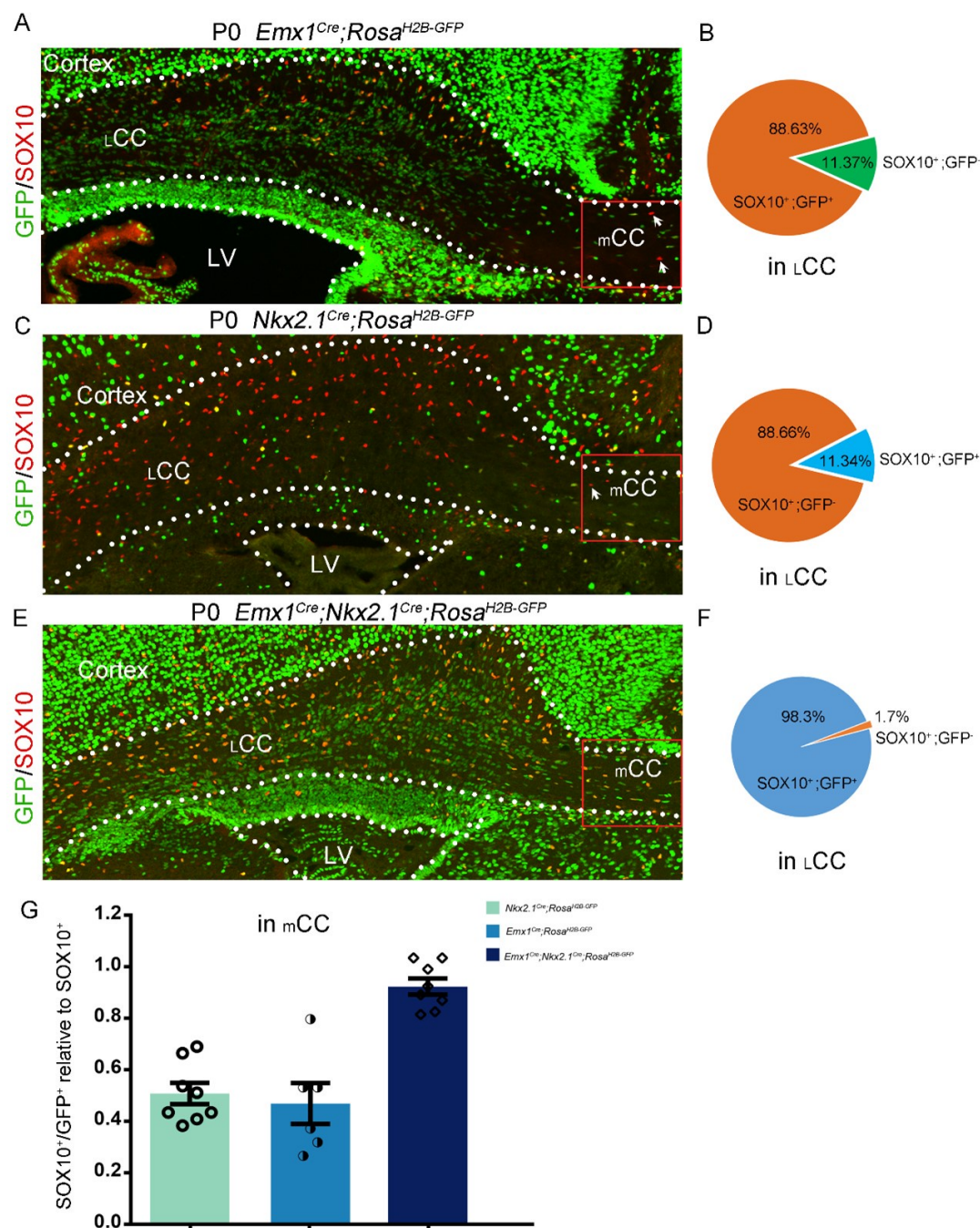
(1) We used the Emx1Cre; RosaH2B-GFP mouse and found that nearly all GSX2+ cells in the cortical SVZ are derived from the Emx1+ lineage at P0 (Please see our new Figure 3-supplement 1A-C).

(2) According to your suggestion, we have cited this paper (Tripathi et al.) in our revised manuscript.

(3) The study conducted by Kessaris et al. (2006) revealed that roughly 50% of cortical oligodendrocytes (OLs) originate from the Gsx2 lineage (LGE/CGE-derived). In contrast, Tripathi et al. (2011) observed that Gsx2-derived OLs contribute only around 20% to the corpus callosum (CC). To investigate the reasons behind these disparate findings, we conducted three experiments. Firstly, using Emx1Cre; RosaH2B-GFP mice, we found that approximately 89% of lateral CC (LCC) OLs originate from the Emx1 lineage, with only around 11% derived from the ventral source (refer to Author response image 1A and B below). Secondly, employing Nkx2-1Cre; RosaH2B-GFP mice, we determined that approximately 11% of LCC OLs originate from the Nkx2.1 lineage (refer to pictures C and D below). Finally, we found that approximately 98.3% of lateral LCC OLs originate from both Emx1 and Nkx2.1 lineages, with only around 1.7% possibly derived from the LGE (see Author response image 1E and F below). Taken together, our results indicate that approximately 89% of LCC OLs originate from the Emx1 lineage, while 11% of LCC OLs are derived from the medial ganglionic eminence (MGE).

It is worth noting that OLs from Emx1 and Nkx2.1 lineages were equally distributed in the medial CC (mCC) (see Author response image 1G below). This finding suggests that MGE-derived OLs exhibit spatial heterogeneity in their distribution within the CC. These results provide evidence that the contribution of the lateral ganglionic eminence (LGE) and caudal ganglionic eminence (CGE) to CC OLs is minimal.

Author response image 1.



Finally, the authors deleted *Olig2* in the MGE and found a dramatic reduction of *PDGFRA*⁺ and *SOX10*⁺ cells in the cortex at E14 and E16 (Figure 4A-F). This further supports their conclusion that, at least at E16, there is no significant contribution of OLs from ventral sources other than the MGE/AEP. This does not exclude the possibility that the LGE/CGE generates OLs for the cortex at later stages. Hence, on its own, this is not completely convincing evidence that the LGE generates no OL lineage cells for the cortex.

There are three reasons why we didn't analyze *Olig2*-NCKO mice after E16.5. 1. The expression of *Nkx2.1*Cre is lower within the dorsal-most region of the MGE than other

Nkx2.1-expressing regions. Even at E15.5, we can still find a small number of OPCs in the lateral cortex. We speculate that these OPCs are derived from dorsal MGE. 2. Considering the possibility of incomplete recombination in Olig2 gene locus, we guess OPCs (Olig2+) in the lateral cortex are derived from MGE. Indeed, we found a few OPCs in the MGE/AEP in the Olig2-NCKO mice (Figure 4F). 3. The recent study (bioRxiv preprint doi: <https://doi.org/10.1101/2024.01.23.576886>) showed that the contribution of LGE/CGE to cortical OPCs is minimal, which further supporting our findings. Taken together, our results provide additional evidence supporting the limited contribution of the LGE/CGE to cortical OPCs (OLs).

Reviewer #2 (Public Review):

Traditional thinking has been that cortical oligodendrocyte progenitor cells (OPCs) arise in the development of the brain from the medial ganglionic eminence (MGE), lateral/caudal ganglionic eminence (LGE/CGE), and cortical radial glial cells (RGCs). Indeed a landmark study demonstrated some time ago that cortical OPCs are generated in three waves, starting with a ventral wave derived from the medial ganglionic eminence (MGE) or the anterior entopeduncular area (AEP) at embryonic day E12.5 (Nkx2.1+ lineage), followed by a second wave of cortical OLs derived from the lateral/caudal ganglionic eminences (LGE/CGE) at E15.5 (Gsx2+/Nkx2.1- lineage), and then a final wave occurring at P0, when OPCs originate from cortical glial progenitor cells (Emx1+ lineage). However, the authors challenge the idea in this paper that cortical progenitors are produced from the LGE. They have found previously that cortical glial progenitor cells were also found to express Gsx2, suggesting this may not have been the best marker for LGE-derived OPCs. They have used fate mapping experiments and lineage analyses to suggest that cortical OPCs do not derive from the LGE.

Strengths:

(1) The data is high quality and very well presented, and experiments are thoughtful and elegant to address the questions being raised.

(2) The authors use two elegant approaches to lineage trace LGE derived cells, namely fate mapping of LGE-derived OPCs by combining IUE (intrauterine electroporation) with a Cre recombinase-dependent IS reporter, and Lineage tracing of LGE-derived OPCs by combining IUE with the PiggyBac transposon system. Both approaches show convincingly that labelled LGE-derived cells that enter the cortex do not express OPC markers, but that those co-labelling with oligodendrocyte markers remain in the striatum.

(3) The authors then use further approaches to confirm their findings. Firstly they lineage trace Emx1-Cre; Nkx2.1-Cre; H2B-GFP mice. Emx1-Cre is expressed in cortical RGCs and Nkx2.1-Cre is specifically expressed in MGE/AEP RGCs. They find that close to 98% of OPCs in the cortex co-label with GFP at later times, suggesting the contribution of OPCs from LGE is minimal.

(4) They use one further approach to strengthen the findings yet further. They cross Nkx2.1-Cre mice with Olig2 F/+ mice to eliminate Olig2 expression in the SVZ/VZ of the MGE/AEP (Figures 4A-B). The generation of MGE/AEP-derived OPCs is inhibited in these Olig2-NCKO conditional mice. They find that the number of cortical progenitors at E16.5 is reduced 10-fold in these mice, suggesting that LGE contribution to cortical OPCs is minimal.

We thank the reviewer for summarizing the strengths of our manuscript.

Weaknesses:

(1) The authors use IUE in experiments mentioned in point 2 of 'Strengths' above (Figures 1 and 2) and claim that the reporter was delivered specifically into LGE VZ at E13.5 using this IUE. It would be nice to see some sort of time course of delivery after IUE to show the reporter is limited to LGE VZ at early times post-IUE.

Thank you very much for your suggestions. Indeed, when using IUE in our system, we occasionally found a small number of electroporated cells in the MGE/AEP VZ. Thus, we can find very few electroporated cells (MGE/AEP-derived) in the cortex and these electroporated cells are neuron (perhaps interneuron).

(2) In the experiments mentioned in point 3 of 'Strengths' (Figure 3), statistical analysis showed that only approximately 2% of OPCs were GFP-negative cells. This 2% could possibly be derived from the LGE/CGE so does not totally rule out that LGE contributes some cortical OPCs.

Thank you for your constructive suggestions. We apologize for any imprecise descriptions. Despite we suspect that this 2% may originate from MGE {Considering the possibility of incomplete recombination in Olig2 gene locus, we guess the OPCs (Olig2+) may be derived from MGE. Indeed, we found a few OPCs in the MGE/AEP in the Olig2-NCKO mice (Figure 4F)} or from the dMGE (The expression of Nkx2.1Cre is lower within the dorsal-most region of the MGE than in other Nkx2.1-expressing regions). Anyway, we have softened the assertion everywhere in our revised manuscript.

(3) In the experiments mentioned in point 4 of 'Strengths' (Figure 4), they do still find cortical OPCs at E16.5 in the Olig2-NCKO conditional mice. It is unclear whether this is due to the recombination efficiency of the CRE enzyme not being 100%, or whether there is some LGE contribution to the cortical OPCs.

This experiment alone may not provide strong evidence to support that LGE do not contribute to the cortical OPCs during development. However, when combining our other results with this result, we can confirm that the contribution of LGE to cortical OPCs is minimal. Furthermore, a recent study reported that LGE/CGE-derived OLs make minimum contributions to the neocortex and corpus callosum, which further supporting the reliability of our conclusion.

We would like to thank the reviewers and editors for their valuable comments and suggestions again.

Impact of Study:

The authors show elegantly and convincingly that the contribution of the LGE to the pool of cortical OPCs is minimal. The title should perhaps be that the LGE contribution is minimal rather than no contribution at all, as they are not able to rule out some small contribution from the LGE. These findings challenge the traditional belief that the LGE contributes to the pool of cortical OPCs. The authors do show that the LGE does produce OPCs, but that they tend to remain in the striatum rather than migrate into the cortex. It is interesting to wonder why their migration patterns may be different from the MGE-derived OPCs which migrate to the cortex. The functional significance of these different sources of OPCs for adult cortex in homeostatic or disease states remains unclear though.

Recommendations for the authors:

Reviewer #1 (Recommendations for The Authors):

(1) Change the title to e.g. 'limited contribution of the LGE to cortical oligodendrocytes'. Alternatively, It might be more useful to highlight where they come from, e.g. "Cortical oligodendrocytes originate predominantly or exclusively from the MGE and cortical VZ"

As suggested, we have changed the old title to the following: The lateral/caudal ganglionic eminence makes a limited contribution to cortical oligodendrocytes

(2) Demonstrate using lineage tracing that GSH2+ cells in the cortex are derived from the Emx1-lineage, e.g. using immunohistochemistry for GSX2 and a reporter in Emx1-Cre mice crossed to a reporter.

In our revised manuscript, we have added a new figure (Figure 3-supplement 1A-C) to demonstrate that the GSX2+ cells in the cortex are derived from the Emx1-lineage.

(3) Make it clear in their discussion that they have not explored the CGE so it is possible that this region generates some OLs.

The Emx1Cre; Nkx2.1Cre; H2B-GFP mice showed that only ~2% cortical OLs are derived from LGE/CGE. Actually, considering the efficiency of Cre enzyme recombination and the relatively low Cre activity in the dMGE of Nkx2.1Cre, the actual contribution of LGE/CGE-derived cortical OLs could be even lower than our current observation. Therefore, our results demonstrate that the LGE/CGE generate very few, possibly even no, OLs for the cortex.

(4) Soften the assertion that the LGE does not generate any OL lineage cells that reach the cortex by e.g. changing the word 'sole' to 'predominant' (line 88) and, elsewhere in the paper, leaving open the possibility that small numbers of LGE-derived OLs might enter the cortex, depending on where exactly one looks.

As suggested, we have softened the assertion everywhere in our manuscript.

(5) Lines 255-260: 'First, the time window during which the MGE generates OLs is very brief, perhaps occurring before MGE neurogenesis. The high level of SHH in the MGE allows for the production of a small population of cortical OPCs around E12.5. Subsequently, multipotent intermediate progenitors begin to express DLX transcription factors resulting in ending the generation of OPCs in the MGE'. What is the evidence that OL genesis precedes neurogenesis? If there is none (as I suspect) then this statement should be removed.

The editors raised a good point. We have no strong evidence to support that OL genesis precedes neurogenesis in MGE, thus, we removed these sentences in our manuscript.

(6) Figure 1E should show quantification of cells as a % of electroporated cells and as a % of PDGFRA+ or OLIG2+ or SOX10+ cells, so that the reader might have a clear view of the extent of labelling.

Done.

(7) Figure 4: This is interesting but incomplete. At E14.5 the authors show the presence of PDGFRA+ cells in the telencephalon. However, at E16.5 they show images only of the dorsal-most region of the cortex. If the LGE/CGE begins to generate OLs for the early cortex, they would be expected to appear near the cortico-striatal boundary, as shown in Kessaris 2006 Fig1g-h. In the current manuscript, the authors do not show these regions,

or the LGE and CGE, in their images. It is essential to show PDGFRA immunolabelling at the cortico-striatal boundary and also in the LGE and CGE at E16.5 in control and Olig2 mutant mice. It is also necessary to extend this analysis to E18.5, perhaps showing PDGFRA+ cells streaming from the cortical VZ/SVZ.

There are three reasons why we didn't analyze *Olig2*-NCKO mice after E16.5. 1. Frankly, the expression of *Nkx2.1*Cre is lower within the dorsal-most region of the MGE than other *Nkx2.1*-expressing regions. Even at E15.5, we can still find a small number of OPCs in the lateral cortex. We guess these OPCs are derived from dMGE. 2. Considering the possibility of incomplete recombination in *Olig2* gene locus, we guess OPCs (*Olig2*+) are derived from MGE. In fact, we found a few OPCs in the MGE/AEP in the *Olig2*-NCKO mice (Figure 4F). 3. The recent study (bioRxiv preprint doi: <https://doi.org/10.1101/2024.01.23.576886>) showed that the contribution of LGE/CGE to cortical OPCs is minimal. Taken together, our results provide additional evidence supporting the limited contribution of the LGE/CGE to cortical OPCs (OLs).

(8) Cite Tripathi et al. (2011) and mention the disparity between the findings of that paper and Kessaris et al. (2006) and possible reasons - see main review above.

Done.

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